Pengyi Xiao

Academic Curriculum Vitae | Last updated: October 26, 2026

Intelligent Manufacturing Engineering, China University of Mining and Technology, Beijing (CUMTB)
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I am seeking MPhil or PhD opportunities starting in Fall 2028.

Research Interests

Research on applying world models to autonomous driving, embodied intelligence, and industrial production, with a focus on prediction, planning, and decision-making in dynamic environments.

Education

China University of Mining and Technology, Beijing Sep. 2024 – Jun. 2028 (expected)
B.Eng. in Intelligent Manufacturing Engineering

Tsinghua University, School of Vehicle and Mobility Sep. 2025 – present
Jointly Supervised Undergraduate Researcher, Academician Li Jun's Research Group
Host advisor: Prof. Huang Chaosheng Home advisor: Prof. Zhang Yunjiu

Shenyang No.31 Senior High School High School
Robotics Club Member
- Built a competition-ready robotics project, sparking an early interest in research; pandemic restrictions prevented national-submission participation.

Research Experience

Researcher, member of Tsinghua VIN Lab, Academician Li Jun's Research Group, Sep. 2025 – present
School of Vehicle and Mobility, Tsinghua University
Role: Jointly Supervised Undergraduate Researcher
- Study time-series forecasting and world-model applications in autonomous driving through trajectory-data processing, literature comparison, experiments, and result analysis.

Project Leader, Undergraduate Research and Innovation Project, Office of Academic Affairs, CUMTB Oct. 2025 – present
"Eye of Demeter – World-Model-Informed Energy Management for an Energy-Efficient Chicken Farm" (Advisor: Prof. Zhang Yankui)
- Lead proposal design, task allocation, advisory coordination, and milestone management; develop a proposed world-model approach to forecasting environmental and energy states for constrained, energy-aware ventilation decisions.

Team Member, Undergraduate Innovation Training Program, CUMTB – Ivy-Inspired Soft Climbing Robot May 2026 – Apr. 2027 (expected)
Advisor: Prof. Li Yan Role: bio-inspired suction-cup design
- Lead bio-inspired adhesion design and multi-surface testing; support integration of pneumatic, SMA, retraction, and visual-perception modules.

Remote Research Intern, Institute of Automation, Chinese Academy of Sciences (Beijing) May 2025 – Jun. 2025
- Applied computer vision techniques to tooth segmentation.

Publications and Research Outputs

Published Conference Paper

1. Zhengxian Chen, Chenggang Zhang, Yuyang Pan, Jingyue Zheng, Fojin Zhou, Peng Ni, Yichong Li, Junyi Wang, Tianyang Gong, Xun Hu, Minhao Gao, Yuxing Wu, Mingyang Li, Pengyi Xiao, Mingqiang Wang, Jun Li. **Energy Management Method for Diesel Hydrogen Hybrid Vehicles and Evaluation of Renewable Energy Applications**. *REM2025, Applied Energy Symposium and Forum*. Co-author; published.

Under Review

2. Jiwei Yang, Zhengxian Chen, Pengyi Xiao, Weihua Pan, Chenggang Zhang, Lei Yang, Wenfeng Guo, Chaosheng Huang, Jun Li. **SDRA: A Spatiotemporally Decoupled RL Agent Enhanced by a Vision-Language Model for Heavy-Duty Vehicle Eco-Driving**. *AAAI 2027*. Co-author; under review.
3. Zhengxian Chen, Han Hai, Rui Li, Pengyi Xiao, Zhengyin Gao, Chaosheng Huang, Jun Li. **The 3D Vehicle Inverted Pendulum Model: A Framework for Comfort-Oriented and Stability-Enhanced Motion Planning in Autonomous Driving**. *ICRA 2027*. Co-author; under review.
4. Jiwei Yang, Zhengxian Chen, Pengyi Xiao, Tianle Wu, Chaosheng Huang, Jun Li. **Auto-JEPA: A Latent World Model of Continuous Intent for End-to-End Autonomous Driving**. *ICLR 2027*. Co-author; under review.
5. Zhengxian Chen, Jinhao Li, Shijia Xu, Pengyi Xiao, Lei Yang, Yanbiao Ma, Chaosheng Huang, Jun Li. **JEPA-UQ: Measuring Uncertainty in Latent World Model Rollouts**. *ICLR 2027*. Co-author; under review.

Works in Progress

- Zhengxian Chen, Pengyi Xiao, Boqi Li, Jiwei Yang, Lechen Hu, Jiapeng Li, Qizhang Wu, Yikai Zhang, Yanbiao Ma, Xiaoshuai Hao, Chaosheng Huang. **World Models for Autonomous Driving**. Review manuscript in preparation; co-author.
- **Patent: Hybrid-Energy Agricultural Fan Emission-Reduction System**. A CUMTB Undergraduate Research and Innovation Project combining vibration energy harvesting, environmental sensing, and world-model-informed ventilation planning.
- **Software Copyright: Agricultural Fan Environment Prediction and Regulation Assistant Software V1.0**. Application in preparation. V1.0 supports offline CSV analysis, standardized ridge-regression forecasts, and threshold-based recommendations; the world-model architecture is a proposed future extension.

Open-Source Contributions

Curator and Contributor, Awesome Autonomous Driving World Model | GitHub Repository

- Curated papers, codebases, and research resources on autonomous-driving world models for a structured, accessible literature map.

Selected Engineering and Innovation Projects

Intelligent Line-Following Car Competition Project – 1st Overall University-Wide Dec. 2024 – Apr. 2025

Role: Hardware and Debugging Lead

- Led vehicle assembly, module integration, debugging, and fault diagnosis for stable competition performance.

Energy/Water Conservation and Mathematical Modeling Competition Projects – First Prize, Beijing Apr. 2025 – present

Role: Team Leader (Overall Coordination)

- Led topic selection, solution refinement, analysis, presentation, and paper preparation.

Biomimetic Blue Morpho Butterfly – 13th Beijing Undergraduate Mechanical Innovation Design Competition, First Prize (Beijing)

Role: Team Member

2nd CUMTB 3D Printing Creative Design Competition Finals – Third Prize Dec. 2025

Role: Team Member

Technical Team Member, *Gaitly* Team – AI Gait Correction and Scenario-Adaptive Modular Baby Walker System, "Challenge Cup" Student Entrepreneurship Competition (UIBE) 2025 – present

- Developed the technical solution; advanced with the team from the UIBE school round.

Awards and Honours

- "Best Crisis Response" Award, Beijing International Model United Nations (BIMUN)
- First Prize, Beijing Energy/Water Conservation and Low-Carbon Emission Reduction Social Practice and Technology Competition
- First Prize (Beijing), National College Students' Mechanical Design Innovation Competition
- Second Prize (Beijing), Higher Education Cup National Undergraduate Mathematical Contest in Modeling
- Third Prize (Beijing), China International College Students' Innovation Competition
- Third Prize (Beijing), 16th "ZhengDa Cup" National College Students' Market Survey and Analysis Competition
- Third Prize (Beijing), National College Students' Financial Challenge Competition
- First Prize, University Line-Following Smart Car Competition (team ranked 1st overall)
- First Prize (College level), Career Planning Competition, School of Mechanical and Electrical Engineering (ranked 1st in college)
- 2025 Summer Social Practice: Advanced Individual; Outstanding Team; First Prize, Outstanding Practice Report
- 2026 Summer Social Practice: Advanced Individual; Outstanding Team; Third Prize, Outstanding Practice Report
- 2024–2025 CUMTB Third-Class Scholarship and Outstanding Student Cadre Recognition (certificate)

Internship Experience

Research Assistant, China Green Intelligence Urban Research Institute — Jul. 2025 – Aug. 2026
Affiliated with the School of Vehicle and Mobility, Tsinghua University
Event Operations Assistant, BMW Brilliance (Huachen BMW) Winter Event — Feb. 2025

Student Leadership

Vice President, University Employment & Entrepreneurship Association — Sep. 2024 – Jun. 2026
President, Model United Nations Association — Sep. 2025 – Jun. 2026
Class Monitor, Intelligent Manufacturing Class 24-1 — Sep. 2024 – Sep. 2025
Team Leader, Social Practice Team to Beijing, School of Mechanical and Electrical Engineering (two consecutive years) — Jul. 2025 – Aug. 2026
Delegation Head, Beijing International MUN (BIMUN), Northeast China Universities MUN (NIMUN), 2025 UIBE MUN — 2024 – 2025
Program Member, 2025 CUMTB (Beijing) New Youth Global Competency Talent Development Program — Jun. 2025 – Mar. 2026

Research-Focused Supplementary Study

Selected self-study to strengthen foundations for research in world models, autonomous driving, and embodied intelligence.

- **CS285, Deep Reinforcement Learning, UC Berkeley** (in progress): Core lectures and assignments, with model-based reinforcement learning as a foundation for world models.
- **6.431x, Probability and Random Processes, MITx** (selected topics): Conditional expectation, martingales, and Markov chains for stochastic modeling and trajectory forecasting.
- **EE364A, Convex Optimization, Stanford University** (selected chapters): Chapters 4–5 and 9–11, emphasizing optimization foundations for MPC and motion planning.
- **Generative Models and Diffusion** (research reading): Yang Song's score-based modeling materials, Lilian Weng's diffusion survey, DDPM, Score-SDE, and Flow Matching.

Language Proficiency

- CET-4 passed; preparing for IELTS and CET-6. Working proficiency in reading academic literature and written research communication.